5 Two point charges, A and B, are separated in a vacuum by a distance of 0.0100 cm, as shown in Fig. 5.1.

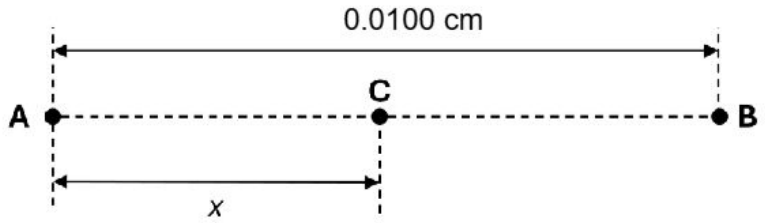


Fig. 5.1

A point C is on the line joining the two charges and is a distance x from charge A.

The variation with distance x of the electric potential V_x at point C is shown in Fig. 5.2.

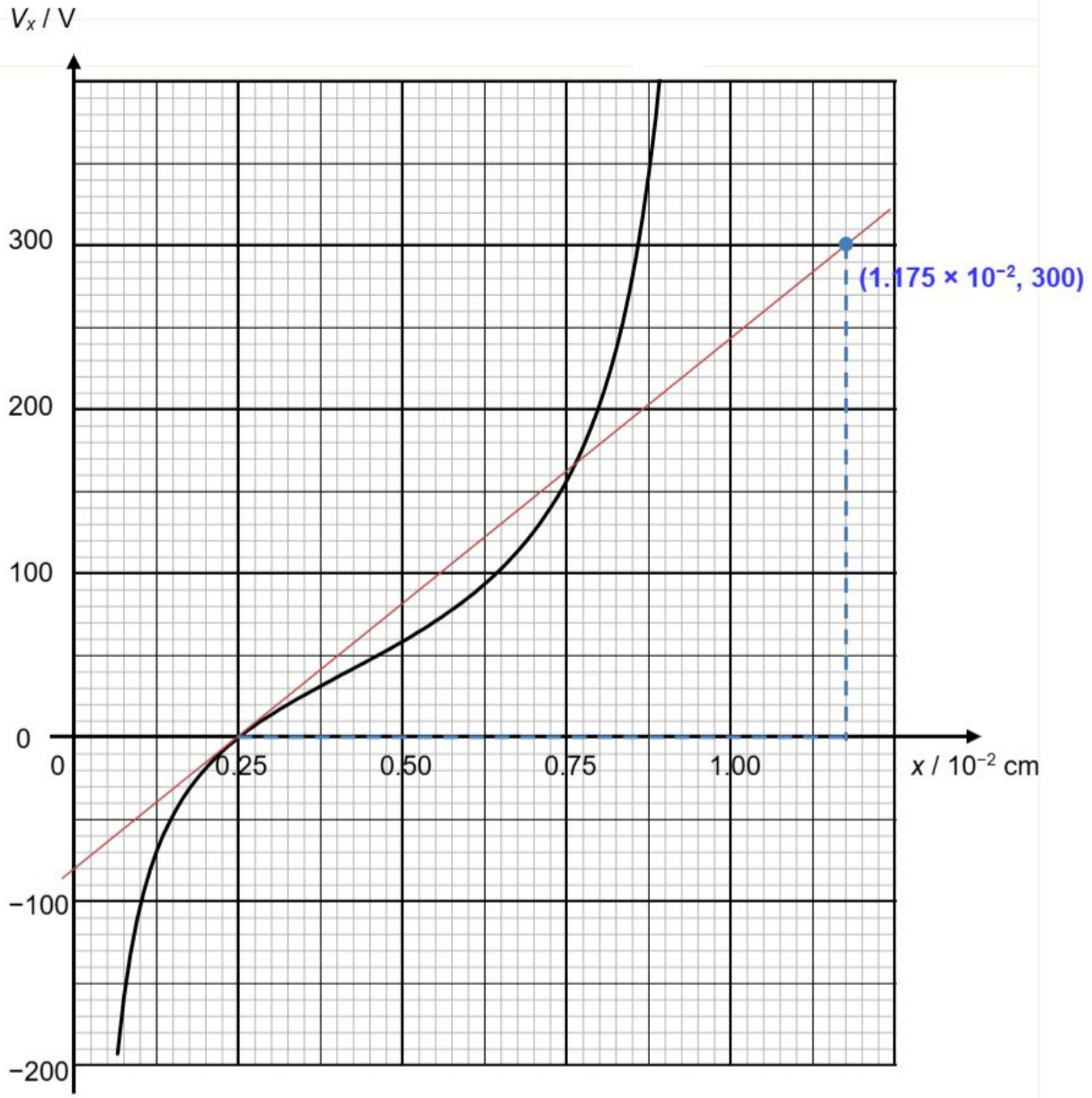


Fig. 5.2

(a) Define electric field strength at a point.

Electric field strength at a point in an electric field is defined as the electric force per unit positive charge exerted on a small test charge placed at that point. [B1]

..... [1]

(b) Explain whether the charges have the same or opposite signs.

The graph shows that the electric potential is zero at a point between the two charges (at x = 0.0025 cm), which is only possible if the two charges are of opposite signs — a positive charge produces a positive potential whilst a negative charge produces a negative potential. [2]

OR

Furthermore, since electric field strength is equal to the negative potential gradient, and the potential decreases continuously from B to A, the resultant electric field points from B towards A at every point between the charges. For this to be the case, the fields due to each individual charge must reinforce rather than cancel, which is only possible if the charges are of opposite signs. [2]

..... [2]

(c) Calculate the electric field strength where $x = 0.25 \times 10^{-2}$ cm.

$$\frac{dV}{dx} = \frac{(300-0)\text{ V}}{(1.175-0)\times10^{-4}\text{ m}} = 2.6\times10^{6}\text{ V m}^{-1} \quad \text{[M1]}$$

$$E = -\frac{dV}{dx}$$

$$= -2.6\times10^{6}\text{ V m}^{-1} \quad \text{[A1]}$$

Negative sign indicates E points in the direction of decreasing V. Therefore, the E has magnitude 2.6×10^{6} V m^{-1} and points in the negative x-direction (leftwards, towards charge A). [B1]

electric field strength = ... V m^{-1}

direction of electric field strength: .....

[3]

(d) An electron is projected with a speed of 7.73×10^6 m s^{-1} from the position where $x = 0.725 \times 10^{-2}$ cm towards point charge A.

Calculate the speed of the electron at the position where $x = 0.175 \times 10^{-2}$ cm.

$\Delta E_k = -\Delta E_p$ {conservation of mechanical energy}

$\Delta E_p = q(V_f - V_i)$ [M1]

$= e\left[(-30) - (140)\right]$

$= \left(-1.60 \times 10^{-19}\right)(-170) = 2.72 \times 10^{-17}$ J

$\therefore \Delta E_k = \frac{1}{2} m_e \left[v_f^2 - v_i^2\right] = -2.72 \times 10^{-17}$ J [M1]

$\Rightarrow v_f = \sqrt{\frac{2\Delta E_k}{m_e} + v_i^2} = \sqrt{\frac{2\left(-2.72 \times 10^{-17}\right)}{9.11 \times 10^{-31}} + \left(7.73 \times 10^6\right)^2}$ **=0.0 ms^{-1}**

[A1]

speed = ... m s^{-1} [3]

[Total: 9]